

§ 182. Proof of the First Auxiliary Lemma for an Odd m .

In the preceding paragraph we made the proof of Theorem I dependent on two auxiliary theorems, whose proof is now still to be sought.

We formulate the first of these auxiliary theorems as follows:

- a) If φ_n is a system of non-zero conjugate functionals of the field Ω_m , of which it is known that

$$\varphi_n^m \quad \text{and} \quad \varphi_n^{-1} \varphi_1^n$$

belong to the principal class, then the functionals φ themselves belong to the principal class.

The theorem will be proved if, for some power φ^k of φ whose exponent is not divisible by q , it can be proved that it belongs to the principal class. For if α, β are numbers in Ω_m and

$$\varphi^m = \varepsilon' \alpha, \quad \varphi^k = \varepsilon'' \beta,$$

then we can determine the rational integers x, y so that

$$mx + ky = 1,$$

and then

$$\varphi = \varphi^{mx} \varphi^{ky} = \varepsilon''' \alpha^x \beta^y;$$

thus, if $\varepsilon', \varepsilon'', \varepsilon'''$ are units, φ itself is associated with a number.

According to the present state of the matter, the proof for an even m is to be carried out by a quite different path from that for an odd m ; we therefore occupy ourselves first with the easier case of an odd m ¹.

The assumptions made in a) about φ_n can, if α, β again denote arbitrary numbers of the field Ω_m and ε unit functionals, be expressed by the two formulas

$$\varphi_n^m = \varepsilon \beta, \tag{1}$$

$$\varphi_n = \varepsilon \alpha \varphi_1^n. \tag{2}$$

We shall now solve the problem step by step.

First let, as before, σ be the prime factor contained in q , which is a number existing in Ω_m , and let σ^k be the power of σ contained in φ ; put

$$\varphi = \sigma^k \psi, \tag{3}$$

where ψ denotes a functional prime to q , which satisfies the same conditions (1), (2) as φ , and if one of the two belongs to the principal class, the same is true of the other. Thus theorem a) needs to be proved only under the assumption that φ is prime to q .

Now let p be a prime number different from q , whose decomposition into prime factors has the expression

$$p = \prod_{\xi} \mathfrak{p}_{\xi}, \tag{4}$$

¹Compare, moreover, the new proof by Hilbert, in which the difference between even and odd m recedes much more (p. 642).

where ξ runs through the number sequence more precisely determined in § 172. Let $\mathfrak{p}_\xi^{k_\xi}$ be the power of $\mathfrak{p}_\xi$ contained in φ , so that, if we set

$$\varphi = \chi \prod_{\xi} \mathfrak{p}_\xi^{k_\xi},$$

then χ is prime to p and is related to no other prime numbers (see the end of § 173) than φ itself. From this there results

$$\varphi_n = \chi_n \prod_{\xi} \mathfrak{p}_{n\xi}^{k_\xi}, \quad (5)$$

and in this we let n again run through the sequence of the numbers ξ . If we further set

$$\psi = \prod_{\xi} \chi_\xi,$$

then ψ too is prime to p and related to no other prime numbers than φ , and from (5), using (4), we obtain

$$\prod_{\xi} \varphi_\xi = \psi p^{\sum k_\xi}, \quad (6)$$

and from this it follows that the functional ψ satisfies the same two conditions (1), (2) as φ .

It is further shown that, if the φ_ξ lie in the principal class, then ψ is also contained in the principal class.

But if $p-1$ is not divisible by q , then we can also conclude the converse. For by (2), φ_ξ is equivalent to φ^ξ , and consequently

$$\prod_{\xi} \varphi_\xi \text{ is equivalent to } \varphi^{\sum \xi}, \quad \text{is equivalent to } \psi,$$

and if ψ is associated with a number, the same holds for $\varphi^{\sum \xi}$. By the theorem § 172, however, if $p-1$ is not divisible by q , $\sum \xi$ is also not divisible by q , and consequently φ itself is contained in the principal class.

This consideration can now be repeated if ψ is still related to further prime numbers which are not congruent to 1 modulo q , and we arrive at the theorem:

Theorem a) is proved in general if it can be proved under the assumption that φ is related only to such prime numbers as are congruent to 1 modulo q .

If we make this assumption about the functionals φ_n , then we can apply Kummer's theorem in the form of § 174, 3.

In what follows we denote by ε unit functionals, whose more precise knowledge is immaterial, and set

$$\Phi_n = \prod_t \varphi_{nt'}^t = \varepsilon \vartheta_n^m, \quad (7)$$

where t runs through the residues of m not divisible by q which are smaller than m , and t' is determined by $tt' \equiv 1 \pmod{m}$. Then ϑ_n is a cyclotomic number in a higher field, whose m th power is contained in Ω_m , and which satisfies the second condition, namely that

$$\vartheta_n^{-1} \vartheta_1^n = \alpha \quad (8)$$

is a number of the field Ω_m .

We now introduce a frequently used sign, understanding by $E(x)$ the greatest integer contained in the real number x , and by (x) the remainder, which is always a proper fraction and, if x itself should be an integer, is to be put equal to zero. Then

$$x = E(x) + (x). \quad (9)$$

If x is an integer, then in this notation

$$m \left(\frac{x}{m} \right)$$

is also an integer, namely the least positive remainder of the division of x by m .

In formula (7), the index of the functionals φ is determined only modulo m . The exponent, however, is reduced to its least remainder modulo m . Thus if in (7) we replace the index t' by $n't'$, then t must be replaced by the remainder of the division of nt by m , that is, [by (9)] by

$$m \left(\frac{nt}{m} \right) = nt - mE \left(\frac{nt}{m} \right),$$

and we can set

$$\Phi_n = \prod_t \varphi_{t'}^{nt - mE(\frac{nt}{m})} = \varepsilon \vartheta_n^m,$$

whereas

$$\Phi_1 = \prod_t \varphi_{t'}^t = \varepsilon \vartheta_1^m.$$

Thus by (8)

$$\Phi_1^n \Phi_n^{-1} = \prod_t \varphi_{t'}^{mE(\frac{nt}{m})} = \varepsilon \alpha^m. \quad (10)$$

Accordingly the quotient

$$\left(\frac{\prod_t \varphi_{t'}^{E(\frac{nt}{m})}}{\alpha} \right)^m$$

is a unit functional, and since it is at the same time the m th power of a functional in Ω_m , the base of this power is also a unit, that is,

$$\prod_t \varphi_{t'}^{E(\frac{nt}{m})} = \varepsilon \alpha,$$

and this product belongs to the principal class. But by our assumption

$$\varphi_{t'} \text{ is equivalent to } \varphi^{t'},$$

and hence

$$\prod_t \varphi_{t'}^{E(\frac{nt}{m})} \text{ is equivalent to } \varphi^{\sum t' E(\frac{nt}{m})},$$

and this holds for every arbitrary n not divisible by q .

If therefore we can prove that n can be so chosen that also the sum

$$S_n = \sum_t t' E \left(\frac{nt}{m} \right)$$

is not divisible by q , then theorem a) is proved.

We first want to make a transformation of the sum S_n , by which the question is led back to the much simpler case in which m is a prime number.

If m is a higher than first power of q , we set

$$m = qm', \quad t = t_1 + qt_2, \quad \begin{array}{l} t_1 = 1, 2, \dots, q-1, \\ t_2 = 0, 1, \dots, m'-1. \end{array}$$

Here m' is still a power of q ; the summation letter t_1 runs through the number sequence $1, 2, \dots, q-1$, and t_2 through the number sequence $0, 1, 2, \dots, m'-1$. We determine t'_1 from the congruence

$$t_1 t'_1 \equiv 1 \pmod{q}$$

and obtain

$$t' \equiv t'_1 \pmod{q}.$$

Then

$$S_n = \sum_t t' E\left(\frac{tn}{m}\right) \equiv \sum_{t_1} t'_1 \sum_{t_2} E\left(\frac{nt_2}{m'} + \frac{nt_1}{m}\right) \pmod{q}. \quad (11)$$

We now assume $n < q$, hence also $nt_1 < m$, so that $nt_1 : m$ is a proper fraction. Then

$$\text{a) } E\left(\frac{nt_2}{m'} + \frac{nt_1}{m}\right) = E\left(\frac{nt_2}{m'}\right),$$

$$\text{b) } \text{or} = E\left(\frac{nt_2}{m'}\right) + 1,$$

and case b) occurs only if between

$$\frac{nt_2}{m'} \quad \text{and} \quad \frac{nt_2}{m'} + \frac{nt_1}{m}$$

there lies an integer. This is the case only if the difference between $nt_2 : m'$ and the next greater integer is smaller than $nt_1 : m$, that is, if

$$E\left(\frac{nt_2}{m'}\right) + 1 - \frac{nt_2}{m'} < \frac{nt_1}{m}. \quad (12)$$

If in the expression on the left side of (12)

$$E\left(\frac{nt_2}{m'}\right) + 1 - \frac{nt_2}{m'} \quad (13)$$

we let t_2 run through all its values, we obtain positive rational fractions not exceeding unity, with denominator m' , no two of which are equal to one another. For if two of the values arising from (13) by substitution of t'_2, t''_2 for t_2 were equal, then $n(t'_2 - t''_2)/m'$ would have to be an integer, which, since n is relatively prime to m' and $t'_2 - t''_2$ is smaller than m' , cannot be. Thus the expressions (13) run in some order through the numerical values

$$\frac{1}{m'}, \frac{2}{m'}, \dots, \frac{m'-1}{m'}, 1, \quad (14)$$

and if a denotes an arbitrary positive fraction, then $E(m'a)$ is the number of numbers of the sequence (14) which are not greater than a . By (12), therefore, the number of values of t_2 for which case b) occurs is equal to $E(nt_1/q)$, and there results

$$\sum_{t_2} E\left(\frac{nt_2}{m'} + \frac{nt_1}{m}\right) = \sum_{t_2} E\left(\frac{nt_2}{m'}\right) + E\left(\frac{nt_1}{q}\right).$$

To form S_n , we multiply this equation by t'_1 and sum. This gives

$$S_n \equiv \sum_{t_1} t'_1 \sum_{t_2} E\left(\frac{nt_2}{m'}\right) + \sum_{t_1} t'_1 E\left(\frac{nt_1}{q}\right) \pmod{q}.$$

Here now $\sum t'_1 = \sum t_1 \equiv 0 \pmod{q}$, because the t_1 pairwise give the sum q , and it follows that

$$S_n \equiv \sum_{t_1} t'_1 E\left(\frac{nt_1}{q}\right) \pmod{q}. \quad (15)$$

The sum standing here on the right side is formed in the same way as S_n itself in the case where m is a prime number. For this case, however, there results

$$\begin{aligned} \frac{nt_1}{q} &= E\left(\frac{nt_1}{q}\right) + \left(\frac{nt_1}{q}\right), \\ \frac{(q-n)t_1}{q} &= E\left(\frac{(q-n)t_1}{q}\right) + \left(\frac{(q-n)t_1}{q}\right), \end{aligned}$$

and from this, by addition,

$$t_1 = E\left(\frac{nt_1}{q}\right) + E\left(\frac{(q-n)t_1}{q}\right) + \left(\frac{nt_1}{q}\right) + \left(\frac{(q-n)t_1}{q}\right),$$

and here the sum of the two positive proper fractions, which nevertheless must be an integer, can have only the value 1. It follows therefore that

$$E\left(\frac{nt_1}{q}\right) + E\left(\frac{(q-n)t_1}{q}\right) = t_1 - 1,$$

and from this, by multiplication with t'_1 and summation over all values of t_1 from 1 to $q-1$,

$$\sum_{t_1} t'_1 E\left(\frac{nt_1}{q}\right) + \sum_{t_1} t'_1 E\left(\frac{(q-n)t_1}{q}\right) \equiv -1 \pmod{q}.$$

Consequently both sums on the left side certainly cannot be divisible by q . Then expression (15) shows that of the two sums S_n and S_{q-n} one certainly is indivisible by q .

This, however, was to be proved.

§ 183. Proof of the Second Auxiliary Lemma for an Odd m .

The second lemma, which by § 181 remains to be proved, is the following:

2. If ε_n is a system of conjugate numerical units of the field Ω_m which satisfies the condition that

$$\varepsilon_n \varepsilon_1^{-n} = \mathcal{E}^m \quad (1)$$

is the m th power of a unit in Ω_m , then the ε_n themselves are m th powers of units in Ω_m , multiplied by a root of unity of the form r^{nk} .

This proof too is quite elementary for the case of an odd m . We therefore first consider this case here.

By § 178, 1., we can represent every unit of the field Ω_m in the form

$$\varepsilon = \pm r^k \mathcal{E}(r), \quad \varepsilon_n = \pm r^{nk} \mathcal{E}(r^n), \quad (2)$$

where $\mathcal{E}(r)$ is a real unit of the field Ω_m . Thus $\mathcal{E}(r)$ satisfies the condition

$$\mathcal{E}(r) = \mathcal{E}(r^{-1}), \quad (3)$$

from which, by (2), follows:

$$\varepsilon_1 \varepsilon_{-1} = \mathcal{E}(r)^2; \quad (4)$$

but from (1), putting $n = -1$, one finds

$$\varepsilon_1 \varepsilon_{-1} = \mathcal{E}^m,$$

that is, the left side of (4) is the m th power of a unit in Ω_m , and hence also

$$\mathcal{E}(r)^2 = \mathcal{E}^m \quad (5)$$

is the m th power of such a unit.

Since now m is odd, the rational integer x can be so determined that

$$2x + m = 1, \quad (6)$$

and from this follows

$$\mathcal{E}(r) = \mathcal{E}(r)^{2x} \mathcal{E}(r)^m.$$

Thus if, according to (5), we set

$$\mathcal{E}^x \mathcal{E}(r) = e(r),$$

then follows

$$\mathcal{E}(r) = e(r)^m. \quad (7)$$

By (7), however, Theorem 2 is proved, and at the same time, for an odd m , the whole Theorem I.

This conclusion too fails for an even m , because then equation (6) is no longer soluble.

§ 184. Preliminary Remarks on the Case of an Even m .

For the case where m is a power of 2, we take an entirely different path, about which some preliminary remarks may first find place here.

We have already seen in § 182 that the first lemma is proved if we can prove that some power of φ , whose exponent is not divisible by q , belongs to the principal class. In doing this, of the two properties of the functional φ only the first is used, namely that the m th power of φ is contained in the principal class.

Now by the general theorems in § 154 we know in every field an exponent h , namely the number of ideal classes, for which φ^h belongs to the principal class whenever φ is any functional in this field.

Thus if we can prove that the class number h in the field Ω_m , when m is a power of 2, is an odd number, then by this alone, without anything further, Lemma 1 is proved.

This is to be the aim of our considerations in the next sections.

With respect to the second lemma the matter is similar. Here, from the assumptions in § 183, 2., one can derive, exactly as above, formula § 183, (5), from which, however, it can only be concluded that $\mathcal{E}(r)$ is the $\frac{1}{2}m$ th power of a real unit $e(r)$.

Thus, assuming the first lemma as proved, formula § 181, (14) gives:

$$(r^n, x_0) = \rho_n \sqrt{e(r^n)} \alpha_n \prod_p (r^{m_2 n}, \eta), \quad (1)$$

where ρ_n is a root of unity of degree m^2 , and α_n is a number in Ω_m , and it would still have to be proved that $e(r^n)$ is the square of a unit in Ω_m .

Since $e(r)$ is a real unit, we have

$$e(r^n) = e(r^{-n}), \quad (2)$$

and we shall further stipulate that the sign of the root is to be so determined (which, by a suitable assumption about ρ_n , we can always assume) that

$$\sqrt{e(r^n)} = \sqrt{e(r^{-n})}. \quad (3)$$

By the assumption, the product $(r^n, x_0)(r^{-n}, x_0)$ is a number of the field Ω_m which is not changed by the substitution (r, r^{-1}) , that is, a real number. Further, by § 174, (5),

$$(r^{m_2 n}, \eta)(r^{-m_2 n}, \eta) = \pm p,$$

so it is likewise real. Similarly $\alpha_n \alpha_{-n}$ is real, and from (1) it follows that

$$\rho_n \rho_{-n} = \frac{(r^n, x_0)(r^{-n}, x_0)}{e(r^n) \alpha_n \alpha_{-n} \prod_p (r^{m_2 n}, \eta)(r^{-m_2 n}, \eta)}$$

is a real number, which, since it is a root of unity, can only be ± 1 .

Now in the equation resulting from this for $n = 1$,

$$(r, x_0)(r^{-1}, x_0) = \pm e(r) \alpha_1 \alpha_{-1} \prod_p (\pm p),$$

we can carry out every substitution (r, r^n) , from which it follows that the sign of $\rho_n \rho_{-n}$ is independent of n , hence that

$$\rho_n \rho_{-n} = \rho_1 \rho_{-1} = \pm 1. \quad (4)$$

Now we further derive from (1):

$$\rho_n \rho_1^{-1} \sqrt{e(r^n)} \sqrt{e(r)}^{-n} = \frac{(r^n, x_0)(r, x_0)^{-n}}{\alpha_n \alpha_1^{-1} \prod_p (r^{m_2 n}, \eta)(r^{m_2}, \eta)^{-n}}, \quad (5)$$

and the right side is a number of the field Ω_m . But the left side shows that it is a unit, and therefore, if $\mathcal{E}(r)$ denotes a real unit, we can set

$$\rho_n \rho_1^{-1} \sqrt{e(r^n)} \sqrt{e(r)}^{-n} = r^k \mathcal{E}(r). \quad (6)$$

If we substitute this value into formula (5) and make the substitution (r, r^{-1}) , there results

$$\rho_{-n} \rho_1^{-1} \sqrt{e(r^n)} \sqrt{e(r)}^{-n} = r^{-k} \mathcal{E}(r), \quad (7)$$

and by multiplication of (6) and (7), with regard to (4),

$$e(r^n) e(r)^{-n} = \mathcal{E}(r)^2. \quad (8)$$

In our considerations on the class number the following theorem will still result:

A unit $e(r)$ of the real field H_m which is positive with all its conjugates is the square of a unit in the field H_m .

Formula (8), however, shows that the unit $\pm e(r)$ has this property, and hence that $e(r)$ (since also $-1 = i^2$ is the square of a unit) is the square of a unit of the field Ω_m .

Thus the question has been reduced to the proof of the two mentioned theorems, which arises as the capstone of a long chain of considerations, but one extremely rich in interesting relations, to which the next sections are to be devoted.

Twenty-third Section. Class Number.

§ 185. Dirichlet's theorem on the units.

The investigations on the number of ideal classes in an algebraic field, which are now to give us the necessary supplements to the proofs of the preceding section, make use of the methods that Dirichlet introduced into number theory.²

They are no longer purely algebraic, insofar as transcendental functions and numbers, such as the natural logarithm and the number π , also occur in them. Use is also made of passages to the limit, as in integral calculus, from which we have already drawn a fine result in § 156.

The investigations with which we first have to deal are quite general, that is, they hold for every algebraic number field, and it offers no advantage for simplicity to restrict them to special fields, such as the cyclotomic fields.

Thus let Ω be a field of the n th degree, and let

$$\Omega_1, \Omega_2, \dots, \Omega_n \tag{1}$$

be the conjugate fields. The irreducible equation of the n th degree whose roots determine these conjugate fields for us will in general have both real and imaginary roots; the latter, however, always occur pairwise, so that to every imaginary field $\Omega^{(i)}$ there belongs another whose numbers are conjugate imaginary to those of $\Omega^{(i)}$, and which therefore arises from $\Omega^{(i)}$ by the substitution $(i, -i)$.

We combine such imaginary pairs into one unit and denote by ν the number of real fields and of pairs of imaginary fields in the sequence (1). If all conjugate fields are real, then $n = \nu$; if they are all imaginary, then $n = 2\nu$.

Except in the case of the rational field, ν is equal to 1 only in the case of an imaginary quadratic field. Otherwise ν is always greater than 1.

The number of imaginary pairs is $n - \nu$, and the number of real fields is $2\nu - n$.

In the determinant by which, in § 145, the square root of the ground number, $\sqrt{\Delta}$, is defined, the substitution $(i, -i)$ interchanges two conjugate imaginary rows at a time, and $\sqrt{\Delta}$ therefore changes its sign precisely when this number is odd, and otherwise not. In the first case $\sqrt{\Delta}$ is imaginary, in the second real. Hence it follows that the ground number Δ of the field Ω is positive or negative according as $n - \nu$ is even or odd.

If from two conjugate imaginary fields we retain only one, there results the sequence of conjugate fields

$$\Omega_1, \Omega_2, \dots, \Omega_\nu,$$

to which we assign a system of signs

$$\delta_1, \delta_2, \dots, \delta_\nu$$

with the convention that $\delta_s = 1$ if Ω_s is real, and $= 2$ if Ω_s is imaginary, so that $\sum \delta_s = n$.

²Dirichlet, *Recherches sur diverses applications de l'analyse infinitésimale à la théorie des nombres*. Crelle's Journal, vols. 19, 21. Dirichlet's Works, vol. I (1839, 1840). (The posthumous papers of Gauss, "De nexu inter multitudinem classium etc." Gauss' Works, vol. II, also belong here.) The application of these principles to cyclotomic numbers was first made by Kummer [Crelle's Journal, vols. 35, 40 (1847, 1850); Liouville's Journal, vol. 16 (1851)]. The general theory of units was likewise founded by Dirichlet [Dirichlet's Works, vol. I, pp. 622, 633, 639 (1840, 1842, 1846)]. The general theory of units and the determination of the class numbers was given by Dedekind: Dirichlet-Dedekind, *Vorlesungen über Zahlentheorie*, 4th ed., § 183 ff. Minkowski, *Geometrie der Zahlen*.

Let η denote a non-zero number of the field Ω and let

$$\eta_1, \eta_2, \dots, \eta_n$$

be the conjugate numbers. To this number system we assign another number system

$$\lambda_1, \lambda_2, \dots, \lambda_\nu,$$

which we define by the equation

$$\lambda_s = \delta_s \log |\eta_s|, \quad (2)$$

where $|\eta_s|$ denotes the absolute value of η_s , and $\log |\eta_s|$ denotes the real natural logarithm of the positive number $|\eta_s|$. If Ω_s is real and the sign $\pm$ is chosen so that $\pm\eta_s$ is positive, then

$$\lambda_s = \log(\pm\eta_s).$$

But if η_s, η'_s is an imaginary pair, then

$$\lambda_s = \log \eta_s \eta'_s,$$

and to η_s and η'_s belongs the same λ_s , so that the number of different λ_s is equal to ν . From this determination one further obtains

$$\lambda_1 + \lambda_2 + \dots + \lambda_\nu = \log N_a(\eta), \quad (3)$$

where, as before, N_a denotes the absolute norm.

We shall call the numbers $\lambda_1, \lambda_2, \dots, \lambda_\nu$ the conjugate logarithms of the number η .

If η is an integer of the field Ω , then its absolute norm is a natural number, which is equal to 1 only when η is a unit; and from (3) it follows that:

1. The sum of the conjugate logarithms of an integer η is positive, and is equal to zero only when η is a unit.

Let $\omega_1, \omega_2, \dots, \omega_n$ denote a basis of the integers in Ω (a minimal basis of Ω , § 145), and let $\omega_{1,s}, \omega_{2,s}, \dots, \omega_{n,s}$, for $s = 1, 2, \dots, n$, denote the conjugate numbers. Then all integers of the field Ω_s can be represented in the form

$$\eta_s = x_1 \omega_{1,s} + x_2 \omega_{2,s} + \dots + x_n \omega_{n,s}, \quad (4)$$

with rational integral coefficients $x_1, x_2, \dots, x_n$. The determinant of this system,

$$\sum \pm \omega_{1,1} \omega_{2,2} \dots \omega_{n,n} = \sqrt{\Delta},$$

is the square root of the discriminant Δ of the field, and hence is always different from zero. Thus the system (4) can be solved for the unknowns $x_1, x_2, \dots, x_n$, and if we now require that the absolute values of the numbers η_s are not to exceed a certain bound, then these solutions give bounds for the rational integers x_i .

This simple remark gives us the following important theorem:

2. In an algebraic field Ω there is only a finite number of integers η of the property that the absolute values of the conjugate numbers η_s lie below a given bound.

We now make use of the general theorem § 155, (25), according to which, if $f(x_1, x_2, \dots, x_n)$ is any positive quadratic form in n variables with determinant D , the variables x_i can be determined as rational integers, not all zero, so that

$$f(x_1, x_2, \dots, x_n) < n \sqrt[n]{0.404D},$$

that is, becomes smaller than a number depending only on the determinant D .

We apply this theorem, as in § 156, to the form

$$f(x_1, x_2, \dots, x_n) = \frac{|\eta_1|^2}{c_1^2} + \frac{|\eta_2|^2}{c_2^2} + \dots + \frac{|\eta_n|^2}{c_n^2},$$

where the η_s are the linear forms (4), and in the case of a real η_s

$$|\eta_s|^2 = (\eta_s)^2,$$

in the case of an imaginary pair η_s, η'_s

$$|\eta_s|^2 = |\eta'_s|^2 = \eta_s \eta'_s.$$

In the latter case $\eta_s \eta'_s$ is the sum of two real squares, and hence, if the c_s are taken to be real, f is a positive form. We further stipulate, for the hitherto arbitrary real quantities c_s , that if η_s, η'_s form a conjugate imaginary pair, then

$$c_s = c'_s \tag{5}$$

is to hold; besides this we subject the c_s to the condition

$$c_1 c_2 \cdots c_n = 1. \tag{6}$$

We form the determinant of the function f , as in § 156, by first regarding it as a form in the variables $\eta_1, \eta_2, \dots, \eta_n$, and the determinant of this form is, because of (6), equal to ± 1 . The determinant of f with respect to the variables x_i is therefore (Vol. I, § 56), apart from the sign, equal to the square of the determinant of the linear forms η_s , that is, to the ground number of the field Ω .

It follows that one can find a number A , independent of the c_s and determined only by the field Ω , of such a kind that, whatever the c_s may be, $f(x_1, x_2, \dots, x_n)$ for integral non-zero x drops below A . Since f is a sum of positive summands, the same holds for each of these summands, and the following theorem is thereby proved:

3. If c_s is any system of real positive numbers satisfying the conditions (5), (6), then there is a real number A , determined by the field Ω alone, such that one can always determine an integer η of the field Ω which, together with its conjugate numbers η_s , satisfies the conditions

$$|\eta_s| < c_s A. \tag{7}$$

Incidentally it follows from this that in every algebraic field Ω , except the rational and the imaginary quadratic field, one can find non-zero integers whose absolute value falls below any given bound.

We shall transform the expression of theorem 3 somewhat by introducing the conjugate logarithms of η . Put $\log A = k$, so that k is a real number determined by Ω , and further

$$c_s = e^{\frac{\gamma_s u}{\delta_s}}, \tag{8}$$

where u may be any real quantity, and where the numbers γ_s satisfy, because of (5) and (6), the condition

$$\gamma_1 + \gamma_2 + \dots + \gamma_\nu = \sum_{s=1}^{\nu} \gamma_s = 0. \tag{9}$$

Since, moreover,

$$|\eta_s| = e^{\frac{\lambda_s}{\delta_s}},$$

it follows from (7) and (8) that

$$\lambda_s - \gamma_s u < k\delta_s. \quad (10)$$

The sum of the ν expressions on the left side of this inequality is, by (9), equal to $\sum \lambda_s$, hence, by theorem 1, is not negative; and we find

$$0 \leq \sum_s (\lambda_s - \gamma_s u) < kn.$$

It follows from this, in view of (10), that a term of this sum, $\lambda_s - \gamma_s u$, cannot be smaller than $-(n - \delta_s)k$, for otherwise the total sum would be negative; and if, for simplicity, we take the lower bound somewhat smaller,

$$-nk\delta_s < \lambda_s - \gamma_s u < k\delta_s. \quad (11)$$

It is thereby proved that for every given system of numbers u, γ_s , provided only that condition (9) is fulfilled, there is an integer η of the field Ω which, with its conjugate numbers, satisfies the limiting conditions (11).

Now let g_s be a system of real quantities of which we demand only that

$$\gamma_1 g_1 + \gamma_2 g_2 + \cdots + \gamma_\nu g_\nu = \alpha$$

be different from zero. By (9) this excludes the case where all the g_s are equal to one another; but if they are not, then for every given system of the g_s there will be infinitely many systems γ_s that satisfy the requirement (9).

If now $k \sum \delta_s g_s = g$ is put, then from (11), by multiplication with g_s and summation, one obtains

$$-ng + \alpha u < \sum g_s \lambda_s < g + \alpha u. \quad (12)$$

According to this limiting condition, keeping g_s and α fixed, we now determine a sequence of integers

$$\eta, \eta', \eta'', \eta''', \dots, \quad (13)$$

by making, for u , a sequence of assumptions

$$u, u', u'', u''', \dots,$$

which are more precisely determined as follows.

First choose u so that

$$-ng + \alpha u = a, \quad g + \alpha u = a'$$

are positive; then put

$$\delta = \frac{a' - a}{\alpha} = \frac{(n+1)g}{\alpha},$$

and put

$$\begin{aligned} u' &= u + \delta, & u'' &= u' + \delta, & u''' &= u'' + \delta, & \dots, \\ a + \alpha\delta &= a', & a' + \alpha\delta &= a'', & a'' + \alpha\delta &= a''', & \dots, \end{aligned}$$

and from (12) we thus obtain, if $\lambda'_s, \lambda''_s, \dots$ are the conjugate logarithms of $\eta', \eta'', \dots$,

$$\begin{aligned} a &< \sum g_s \lambda_s < a', \\ a' &< \sum g_s \lambda'_s < a'', \\ a'' &< \sum g_s \lambda''_s < a''', \\ &\dots\dots\dots, \end{aligned} \quad (14)$$

and the sequence of these inequalities can be continued without bound.

Moreover all these numbers $\eta, \eta', \eta'', \dots$ have, by (6) and (7), the property that their absolute norms $N, N', N'', \dots$ remain below a fixed finite bound e^{nk} .

The number of possible values of $N, N', N'', \dots$ is therefore finite, certainly not greater than the greatest integer contained in e^{nk} . Likewise the number of all possible residues of the numbers $x_1, x_2, \dots, x_n$ [in (4)] modulo all the moduli $N, N', N'', \dots$ is finite; and from this it follows that, if we continue the sequence of numbers (13) far enough, two distinct numbers $\eta^{(h)}, \eta^{(k)}$ must occur in the sequence for which $N^{(h)} = N^{(k)}$ and the residues of $x_1, x_2, \dots, x_n$ modulo $N^{(h)}$ are exactly the same.

But from (14), if $h > k$ is assumed, it follows that

$$\sum g_s(\lambda_s^{(h)} - \lambda_s^{(k)}) > 0. \quad (15)$$

If N is the absolute norm of these two numbers $\eta^{(h)}, \eta^{(k)}$, then, because of the agreement of the residues of the x , the difference $\eta^{(h)} - \eta^{(k)}$ is divisible by N . On the other hand, by § 138 (13), the number N is divisible by $\eta^{(h)}$, and consequently $\eta^{(h)} - \eta^{(k)}$ is also divisible by $\eta^{(h)}$. It follows that $\eta^{(k)}$ is also divisible by $\eta^{(h)}$, and likewise $\eta^{(h)}$ by $\eta^{(k)}$. The two numbers are therefore associated, and if we put

$$\frac{\eta^{(h)}}{\eta^{(k)}} = \varepsilon,$$

then ε is a unit of the field Ω .

Moreover, if, denoting by $|\varepsilon_s|$ the absolute value of ε_s , we put

$$\delta_s \log |\varepsilon_s| = l_s, \quad (16)$$

then from (15) we obtain

$$g_1 l_1 + g_2 l_2 + \dots + g_\nu l_\nu > 0. \quad (17)$$

This proves the theorem:

4. If $g_1, g_2, \dots, g_\nu$ is any system of real numbers which are not all equal to one another, then in every algebraic field Ω one can determine a unit ε such that the sum

$$g_1 l_1 + g_2 l_2 + \dots + g_\nu l_\nu$$

is different from zero, where l_s is the system of conjugate logarithms of ε .

This important theorem, which is the foundation for what follows, is due, though in a somewhat changed form, to Dirichlet. This formulation was given by Minkowski.

§ 186. Systems of independent units and exponent systems of the units.

If in the theorem just proved we put $g_1 = 1, g_2 = 0, \dots, g_\nu = 0$, which is allowed if from now on we assume $\nu > 1$, it follows that in Ω there is a unit ε for which one of the conjugate logarithms, l_1 , is different from zero. Since every power of ε has the same property, there are also infinitely many such units.

This theorem now permits a very important generalization:

5. If $s \leq \nu - 1$, then in the field Ω one can determine a system of units with the associated conjugate logarithms

$$\begin{aligned} \varepsilon_1 : & l_{1,1}, l_{1,2}, \dots, l_{1,\nu}, \\ \varepsilon_2 : & l_{2,1}, l_{2,2}, \dots, l_{2,\nu}, \\ & \dots\dots\dots \\ \varepsilon_s : & l_{s,1}, l_{s,2}, \dots, l_{s,\nu}, \end{aligned} \quad (1)$$

in such a way that any one of the s -rowed determinants of the matrix of the $l_{h,k}$, for instance

$$L_s = \sum \pm l_{1,1} l_{2,2} \cdots l_{s,s},$$

is different from zero.

For $s = 1$ the stated theorem is correct by the remark made at the beginning. We therefore assume that it has been proved for $s - 1$ and derive it generally by complete induction. For this purpose we arrange the determinant L_s according to the elements of the last row, and write it thus:

$$L_s = g_1 l_{s,1} + g_2 l_{s,2} + \cdots + g_s l_{s,s}, \quad (2)$$

where $g_s = L_{s-1}$ and hence, by the assumption made, can be taken to be different from zero. If we substitute these values of g in the formula of theorem 4, § 185, taking $g_{s+1} = 0, \dots, g_\nu = 0$, while g_s is not equal to zero, then, as long as $s < \nu$, certainly not all $g_1, g_2, \dots, g_\nu$ are equal to one another; and it follows that ε_s can be so chosen that L_s is different from zero, as was to be proved.

This mode of inference cannot be extended to $s = \nu$, and the determinant L_ν must indeed always be zero, since for every unit $\sum_1^\nu l_s$ vanishes.

For the case $s = \nu - 1$ we shall now denote the determinant L_s by

$$L_{\nu-1} = L(\varepsilon_1, \varepsilon_2, \dots, \varepsilon_{\nu-1}), \quad (3)$$

and call a system of units $\varepsilon_1, \varepsilon_2, \dots, \varepsilon_{\nu-1}$ for which this determinant is different from zero a system of independent units.

The absolute value L of the determinant $L(\varepsilon_1, \varepsilon_2, \dots, \varepsilon_{\nu-1})$ is called the regulator of the system $\varepsilon_1, \varepsilon_2, \dots, \varepsilon_{\nu-1}$.³

If in the determinant

$$\begin{vmatrix} l_{1,1} & l_{1,2} & \cdots & l_{1,\nu} \\ \cdots & \cdots & \cdots & \cdots \\ l_{\nu-1,1} & l_{\nu-1,2} & \cdots & l_{\nu-1,\nu} \\ x_1 & x_2 & \cdots & x_\nu \end{vmatrix}, \quad (4)$$

in which $x_1, x_2, \dots, x_\nu$ are arbitrary quantities, we add all columns to the last one, then, with regard to the relations $\sum_1^\nu l_{s,i} = 0$, we obtain its value equal to

$$\pm(x_1 + x_2 + \cdots + x_\nu)L.$$

If, therefore, we take all the x except for an arbitrary one to be $= 0$, and the remaining one to be $= 1$, then from (3) there results one of the $(\nu - 1)$ -rowed determinants of the matrix

$$\begin{vmatrix} l_{1,1} & \cdots & l_{1,\nu} \\ \cdots & \cdots & \cdots \\ l_{\nu-1,1} & \cdots & l_{\nu-1,\nu} \end{vmatrix}$$

³Dirichlet-Dedekind, *Vorlesungen über Zahlentheorie*, 4th ed., § 183.

and all these determinants therefore have, apart from the sign, the same value. It is therefore immaterial which one of the ν conjugate values is omitted in forming the regulator L .

If $\varepsilon_1, \varepsilon_2, \dots, \varepsilon_{\nu-1}$ is an independent system of units, and if $\lambda_1, \lambda_2, \dots, \lambda_\nu$ are the conjugate logarithms of any unit ε in Ω , then the numbers $\xi_1, \xi_2, \dots, \xi_{\nu-1}$ can always and uniquely be determined so that

$$\begin{aligned} \xi_1 l_{1,1} + \xi_2 l_{2,1} + \dots + \xi_{\nu-1} l_{\nu-1,1} &= \lambda_1, \\ \xi_1 l_{1,2} + \xi_2 l_{2,2} + \dots + \xi_{\nu-1} l_{\nu-1,2} &= \lambda_2, \\ &\dots\dots\dots \\ \xi_1 l_{1,\nu} + \xi_2 l_{2,\nu} + \dots + \xi_{\nu-1} l_{\nu-1,\nu} &= \lambda_\nu \end{aligned} \tag{5}$$

holds. For the first $\nu - 1$ of these equations form a system of linear equations with non-vanishing determinant, and the ν th equation follows from the others because the sum of the conjugate logarithms of every unit vanishes.

We call the system of numbers $\xi_1, \xi_2, \dots, \xi_{\nu-1}$ the exponent system of the unit ε with respect to the system $\varepsilon_1, \varepsilon_2, \dots, \varepsilon_{\nu-1}$, and it remains to prove:

6. That the exponents of every unit ε are rational numbers whose denominators do not exceed a certain finite bound, and which therefore can all be assumed to have a common denominator depending only on the system ε_i .

To see this, one must consider the following:

1) If we regard the quantities $\xi_1, \xi_2, \dots, \xi_{\nu-1}$ as variables and let each of them, independently of the others, run from 0 to 1, then the left sides of (5) remain enclosed within finite bounds depending only on the units $\varepsilon_1, \varepsilon_2, \dots, \varepsilon_{\nu-1}$. Within these bounds, therefore, the conjugate logarithms λ of the unit ε must also remain, as long as the exponents ξ_i remain restricted to non-negative proper fractional values; and hence, from the definition of the conjugate logarithms, one obtains an upper bound for ε and its conjugates.

But by theorem 2, § 185, there are in Ω only finitely many integers, and therefore still more only finitely many units, which in absolute value, with all their conjugates, lie below a finite bound.

If we now call a unit whose exponents lie between the bounds 0 and 1, including the lower and excluding the upper bound, a unit reduced with respect to the system $\varepsilon_1, \varepsilon_2, \dots, \varepsilon_{\nu-1}$, then we have the theorem:

There are only finitely many units in Ω which are reduced with respect to an independent system $\varepsilon_1, \varepsilon_2, \dots, \varepsilon_{\nu-1}$.

2) The units reproduce themselves by multiplication and division. If

$$\begin{aligned} \xi'_1, \xi'_2, \dots, \xi'_{\nu-1}, \\ \xi''_1, \xi''_2, \dots, \xi''_{\nu-1} \end{aligned}$$

are the exponents of two units $\varepsilon', \varepsilon''$, then the sums and the differences

$$\xi'_1 \pm \xi''_1, \xi'_2 \pm \xi''_2, \dots, \xi'_{\nu-1} \pm \xi''_{\nu-1}$$

are the exponents of the units $\varepsilon'\varepsilon''$ and $\varepsilon' : \varepsilon''$.

This follows immediately from the theorem that the logarithm of a product or of a quotient is equal to the sum or the difference of the logarithms of the two components.

3) The units $\varepsilon_1, \varepsilon_2, \dots$ themselves have the exponents

$$1, 0, \dots, 0; \quad 0, 1, \dots, 0; \quad \dots,$$

and from 2) it follows that every system of integers, substituted for $\xi_1, \xi_2, \dots, \xi_{\nu-1}$, gives the exponent system of a unit which can be formed by multiplication of powers of the $\varepsilon_1, \varepsilon_2, \dots$

4) From 2) and 3) it follows that an exponent system $\xi_1, \xi_2, \dots, \xi_{\nu-1}$ of a unit remains an exponent system if all its elements are multiplied by one and the same integer, and that it also retains this character when its elements ξ_i are increased or decreased by arbitrary integers. Thus, if we use the sign (x) in the sense of § 182, that it denotes the excess of the number x over the greatest integer contained in x , the following is obtained:

If $\xi_1, \xi_2, \dots, \xi_{\nu-1}$ is the exponent system of a unit, and m is an integer, then also

$$(m\xi_1), (m\xi_2), \dots, (m\xi_{\nu-1}) \quad (6)$$

is the exponent system of a unit.

Now the quantities $(m\xi_i)$, when they are not zero, are positive proper fractions; and by 1), if the sequence of integers is continued sufficiently far, it must therefore occur that for two distinct values m', m'' of m the number sequence (6) agrees. Since accordingly $m'\xi_i, m''\xi_i$ have the same excess over an integer, $(m' - m'')\xi_i$ is itself an integer; and so there is a rational integer m , at most equal to the number of exponent systems of reduced units, with the property that

$$m\xi_1, m\xi_2, \dots, m\xi_{\nu-1}$$

become rational integers.

This number m may change when the unit ε is changed. But since all possible values of this number lie below a finite bound, there is a finite number in which all these values of m are contained and which itself can be taken for m . Hence there is a certain positive integer m which can be taken as denominator of all rational numbers which can occur as exponents of units. This proves theorem 6.

§ 187. Fundamental systems of units.

We now choose, instead of the system of independent units ε_i , another system whose conjugate logarithms may be

$$\lambda_{i,1}, \lambda_{i,2}, \dots, \lambda_{i,\nu} \quad i = 1, 2, \dots, \nu - 1. \quad (1)$$

By § 186, (5),

$$\lambda_{i,k} = \xi_{1,i}l_{1,k} + \xi_{2,i}l_{2,k} + \dots + \xi_{\nu-1,i}l_{\nu-1,k}, \quad (2)$$

$$i = 1, 2, \dots, \nu - 1; \quad k = 1, 2, \dots, \nu,$$

where $\xi_{1,i}, \xi_{2,i}, \dots, \xi_{\nu-1,i}$ denote the exponents of one of the new units with respect to the system $\varepsilon_1, \varepsilon_2, \dots, \varepsilon_{\nu-1}$, which we may assume to be rational numbers with the common denominator m .

But by the multiplication theorem for determinants,

$$\begin{vmatrix} \lambda_{1,1} & \dots & \lambda_{\nu-1,1} \\ \dots & \dots & \dots \\ \lambda_{1,\nu-1} & \dots & \lambda_{\nu-1,\nu-1} \end{vmatrix} = \begin{vmatrix} \xi_{1,1} & \dots & \xi_{1,\nu-1} \\ \dots & \dots & \dots \\ \xi_{\nu-1,1} & \dots & \xi_{\nu-1,\nu-1} \end{vmatrix} \begin{vmatrix} l_{1,1} & \dots & l_{\nu-1,1} \\ \dots & \dots & \dots \\ l_{1,\nu-1} & \dots & l_{\nu-1,\nu-1} \end{vmatrix}, \quad (3)$$

or, if we put

$$\Lambda_{\nu-1} = \sum \pm \lambda_{1,1} \lambda_{2,2} \dots \lambda_{\nu-1,\nu-1}$$

and observe that the determinant

$$\sum \pm \xi_{1,1} \xi_{2,2} \dots \xi_{\nu-1,\nu-1}$$

is a rational number with denominator $m^{\nu-1}$,

$$\Lambda_{\nu-1} = \frac{a}{m^{\nu-1}} L(\varepsilon_1, \varepsilon_2, \dots, \varepsilon_{\nu-1}), \quad (4)$$

where a is a rational integer.

It follows that the newly introduced units form an independent system if and only if the determinant of the $\xi_{i,k}$, that is, the number a , is different from zero.

Now among a finite or infinite number of non-zero rational fractions with the same denominator there is always one of least absolute value, and consequently, by (4), we can choose the new system of independent units so that its regulator $\pm\Lambda_{\nu-1}$ becomes as small as possible. Such a system we call a fundamental system of units; and the minimum value of the regulator itself, that is, the regulator of a fundamental system, we call the regulator of the field. We now assume that our system $\varepsilon_1, \varepsilon_2, \dots, \varepsilon_{\nu-1}$ itself is a fundamental system.

Under this assumption it can be proved that all exponents of units must be integers.

Indeed, if we assume that there exists a unit ε whose exponents determined by the formulae § 186, (5) are not all integers, then by § 186, 4) there is also a unit ε_0 whose exponents are proper fractions and do not all vanish.

If by $\lambda_1, \lambda_2, \dots, \lambda_{\nu-1}$ in the formulae (5), § 186, we understand the system of conjugate logarithms of ε_0 , then

$$L(\varepsilon_0, \varepsilon_2, \dots, \varepsilon_{\nu-1}) = \xi_1 L(\varepsilon_1, \varepsilon_2, \dots, \varepsilon_{\nu-1}),$$

and if we therefore assume, which is plainly no restriction of generality, that ξ_1 is a non-zero proper fraction, then this formula contradicts the assumption that

$$\varepsilon_1, \varepsilon_2, \dots, \varepsilon_{\nu-1}$$

is a fundamental system, because the determinant $L_{\nu-1}$ would be made smaller if ε_1 were replaced by ε_0 .

Once we have a fundamental system, we can take any system of rational integers $\xi_1, \xi_2, \dots, \xi_{\nu-1}$ as an exponent system and form a unit ε with these exponents:

$$\varepsilon = \varepsilon_1^{\xi_1} \varepsilon_2^{\xi_2} \cdots \varepsilon_{\nu-1}^{\xi_{\nu-1}}.$$

It remains to answer the question how far a unit is determined by its exponent system.

Suppose that $\varepsilon', \varepsilon''$ are two units with the same exponent system. Then the exponent system of the unit $\varepsilon' : \varepsilon'' = \rho$ consists entirely of zeros, and consequently the conjugate logarithms of ρ are all equal to zero. Therefore ρ is an integer with the property that the absolute values of all numbers conjugate with ρ are equal to 1, and hence, as proved in § 175, 2, it is a root of unity. Thus the following theorem is proved:

1. There is in the field Ω a system of $\nu - 1$ fundamental units

$$\varepsilon_1, \varepsilon_2, \dots, \varepsilon_{\nu-1}, \tag{5}$$

having the property that in the form

$$\varepsilon = \rho \varepsilon_1^{\xi_1} \varepsilon_2^{\xi_2} \cdots \varepsilon_{\nu-1}^{\xi_{\nu-1}} \tag{6}$$

all units of the field, each only once, are contained, when $\xi_1, \xi_2, \dots, \xi_{\nu-1}$ run through all rational integers and ρ runs through all roots of unity present in Ω .

It is now easy to derive all other fundamental systems from one fundamental system, by putting in (6), for the exponents, $\nu - 1$ different systems of integers $\xi_{i,k}$ whose determinant is ± 1 . For if one puts

$$\varepsilon'_i = \rho_i \varepsilon_1^{\xi_{1,i}} \varepsilon_2^{\xi_{2,i}} \cdots \varepsilon_{\nu-1}^{\xi_{\nu-1,i}},$$

then from (3) it follows that

$$L(\varepsilon'_1, \varepsilon'_2, \dots, \varepsilon'_{\nu-1}) = \pm L(\varepsilon_1, \varepsilon_2, \dots, \varepsilon_{\nu-1}).$$

Thus both determinants have the minimum value, and both systems are fundamental systems.

The case $\nu = 1$, which we excluded above, occurs, apart from the case of the rational field, in which only the two units ± 1 exist, in the case of the imaginary quadratic field. In this case, in the twentieth section, we represented the integers of the field in the form

$$\frac{x + y\sqrt{-m}}{2},$$

where $-m$ denotes a fundamental discriminant, so that $m \equiv 0$ or $\equiv 3 \pmod{4}$, and x, y are both even or both odd. We obtain the units if we solve the equation

$$x^2 + my^2 = 4$$

in all possible ways in rational integers. But this equation, if $m > 4$, has only the two solutions $x = \pm 2, y = 0$, and thus in these cases, as in the field of rational numbers, there are only the two units ± 1 . If $m = 4$, one finds the four units $\pm 1, \pm i$, and finally if $m = 3$, the six units

$$\pm 1, \quad \pm \frac{-1 + i\sqrt{3}}{2}, \quad \pm \frac{-1 - i\sqrt{3}}{2}.$$

In the next simplest case, $n = 2, \nu = 2$, that is, in the real quadratic field, the theory of units coincides with the theory of Pell's equation treated in § 127 of the first volume.